The Essential of ML: Chapter 4

Discriminative and generative classification cut down to what the professor asks: two routes to the posterior, logistic regression, LDA and QDA, naive Bayes, prior shift, and the confident-but-unfamiliar input

Fundamentals of Machine Learning

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How to read this document

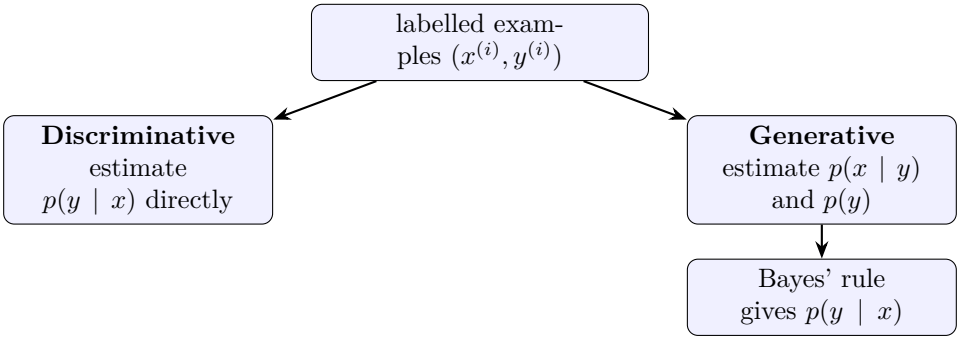
This is chapter 4 reduced to what Tutorial 1 (Problems 2(d), 6 and 7) and the review sheets actually test. Six sections in concept order, every term defined before it is used, each closing with a *Tutorial drill* box whose questions are the professor's (cited) or written in his style, answered the way an expert answers: claim, mechanism, number or derivation, consequence. Left out on purpose: deep networks, non-Gaussian feature models, language-model classification heads, the Lab 2 description; the full chapter has them. Numbers to carry: $\sigma(0) = 0.5$, $\sigma(1.2) = 0.769$, $\ln 2 = 0.693$, $\ln 3 = 1.099$, $-\ln 0.9 = 0.105$, $-\ln 0.1 = 2.303$.

1 Two Routes to the Posterior

Posterior, prior, class-conditional, and the decision that uses them

Let $Y \in \{1, \dots, C\}$ be the class and $X \in \mathbb{R}^d$ the feature vector. The **posterior** $p(y = c \mid x)$ is the probability of class c once x is seen; the **prior** $\pi_c = p(y = c)$ is how frequent class c is before any input is seen; the **class-conditional** $p(x \mid y = c)$ is the distribution of inputs that class c produces (a density or a mass function). Chapter 2 settled what to do *if the posterior is known*: under zero–one loss choose $f^*(x) = \arg\max_c p(y = c \mid x)$; with costs $C_{\text{FP}}, C_{\text{FN}}$ declare class 1 iff $p(y = 1 \mid x) > C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}})$. The rule is one line; the whole difficulty is that the posterior is *unknown* and must be estimated from $\mathcal{D}_N = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$.

There are exactly two ways to get a posterior out of labelled data, and the chapter is their comparison. A **discriminative** model writes down a parametric family $p_\theta(y = c \mid x)$ and fits θ so that it predicts the labels well: it learns *where the label changes* as a function of x and never models x itself. A **generative** classifier writes down the prior π_c and a class-conditional $p_\theta(x \mid y = c)$ for every class, fits them, and obtains the posterior by Bayes' rule: it learns *what each class looks like*, then asks which class explains this x best.



Bayes' rule for classification, and its odds form

Under the generative model the joint is $p_\phi(x, y = c) = \pi_c\, p_\theta(x \mid y = c)$, so by the definition of conditional probability and the sum over the exhaustive classes,

$$p_\phi(y = c \mid x) = \frac{p_\phi(x, y = c)}{p(x)} = \frac{\pi_c\, p_\theta(x \mid y = c)}{\sum_{k=1}^{C} \pi_k\, p_\theta(x \mid y = k)}.$$

The denominator is the same for every c, so classification compares the **prior-weighted likelihoods** $\pi_c\, p_\theta(x \mid y = c)$ only. For two classes, divide the two numerators and the denominator cancels:

$$\underbrace{\frac{p(y = 1 \mid x)}{p(y = 0 \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{\pi_1}{\pi_0}}_{\text{prior odds}} \times \underbrace{\frac{p(x \mid y = 1)}{p(x \mid y = 0)}}_{\text{likelihood ratio } \Lambda(x)}.$$

"Posterior odds equal prior odds times likelihood ratio" drives §3, §4 and §6.

Key insight. Both routes end in the same object, a softmax of class scores. A discriminative model fits the scores directly; a generative model builds them as $\log \pi_c + \log p_\theta(x \mid y = c)$. Everything that separates the routes is *where the scores come from and what had to be assumed to get them.*

Tutorial drill

Tutorial 1, Q1(c): *Explain why a posterior estimate $\hat{p}_\theta(y = 1 \mid x)$ and the alert decision are not the same object.* The posterior is a number in $(0, 1)$ produced by the model from the data; the decision is an action chosen by comparing it with a threshold set by costs. The same posterior 0.35 is "benign" at threshold 0.5 and "attack" at threshold 0.2 ($C_{\text{FP}} = 2$, $C_{\text{FN}} = 8$). Changing the costs changes the action without touching the model; retraining changes the posterior without touching the costs. One probability model supports several policies, and no policy repairs a bad probability.

New: *Priors* 0.99/0.01*; a flow has likelihoods* 0.001 *(benign) and* 0.20 *(attack). Classify it, give the posterior, and say in one sentence why the* 1% *prior does not settle the matter.* Attack: prior-weighted likelihoods $0.99 \times 0.001 = 0.00099$ against $0.01 \times 0.20 = 0.00200$, posterior $0.002/0.00299 = 0.67$. In odds: prior odds $1/99$ times likelihood ratio 200 gives 2.02. The prior is one factor of the posterior odds, the evidence the other, and 200 beats 99; a prior settles the matter only when the input carries no evidence ($\Lambda = 1$).

2 The Discriminative Route: Logistic Regression and Softmax

Score, sigmoid, logistic regression

A discriminative model first produces a real-valued **score** (logit) $s_\theta(x)$, then turns it into a probability, then into an action: $s_\theta(x) \to p_\theta(y \mid x) \to \hat{y}(x)$. For two classes the **sigmoid** $\sigma(s) = 1/(1 + e^{-s})$ does the conversion, $p_\theta(y = 1 \mid x) = \sigma(s_\theta(x))$. **Logistic regression** is the linear score $s_\theta(x) = w^\top x + b$, $\theta = (w, b)$. Facts, all used below: (i) $0 < \sigma < 1$; (ii) $\sigma(-s) = 1 - \sigma(s)$, since $1 - \frac{1}{1+e^{-s}} = \frac{e^{-s}}{1+e^{-s}} = \frac{1}{e^{s}+1}$; (iii) σ is strictly increasing; (iv) $\sigma'(s) = \sigma(s)(1 - \sigma(s))$; (v) for $t \in (0, 1)$, $\sigma(s) > t \iff s > \log \frac{t}{1-t}$ (invert: $e^{-s} < \frac{1-t}{t}$). Values: $\sigma(0) = 0.5$, $\sigma(1.2) = 0.769$, $\sigma(2) = 0.881$.

The log-odds are the score; the boundary is a hyperplane that costs only slide

From $p = \sigma(s)$ and (ii), $\frac{p}{1-p} = \frac{\sigma(s)}{\sigma(-s)} = e^s$, hence

$$\log \frac{p_\theta(y = 1 \mid x)}{p_\theta(y = 0 \mid x)} = s_\theta(x) = w^\top x + b.$$

Logistic regression is the model "log-odds linear in x": one unit more of x_j adds w_j to the log-odds, i.e. multiplies the odds by e^{w_j}, whatever the other features. **Decision at equal costs:** $p > \frac{1}{2} \iff s > 0$, so the boundary is the hyperplane $w^\top x + b = 0$. **Decision under costs:** the threshold $t = C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}})$ on p becomes, by (v), the threshold $s > \log \frac{t}{1-t} = \log \frac{C_{\text{FP}}}{C_{\text{FN}}}$ on the score: a *parallel* hyperplane $w^\top x + b = \log(C_{\text{FP}}/C_{\text{FN}})$. The score family fixes the geometry (linear score, flat boundary; curved score g_θ, curved boundary $g_\theta(x) = 0$); the costs only translate it. Expensive misses ($C_{\text{FN}} = 10$, $C_{\text{FP}} = 1$) move it to $s = \log 0.1 = -2.3$: alarm already at a mildly negative score.

Softmax for C classes

With one score per class, $p_\theta(y = c \mid x) = \text{softmax}(s_\theta(x))_c = \dfrac{e^{s_{\theta,c}(x)}}{\sum_{k=1}^{C} e^{s_{\theta,k}(x)}}$. Three facts: the ordering of probabilities is the ordering of scores, so $\arg\max_c p = \arg\max_c s_c$; adding the same constant a to every score multiplies numerator and denominator by e^a and changes nothing, so only $C - 1$ score *contrasts* are identifiable (subtract the largest score before exponentiating by hand); with $C = 2$, $\frac{e^{s_1}}{e^{s_1}+e^{s_2}} = \frac{1}{1+e^{s_2-s_1}} = \sigma(s_1 - s_2)$. Scores $(2, 0, -1)$: exponentials $(7.39, 1, 0.37)$, total 8.76, probabilities $(0.84, 0.11, 0.04)$.

Cross-entropy is conditional maximum likelihood; its gradient is the error

The **cross-entropy** of example i penalises the probability given to the true class, $\ell_{\text{CE}}^{(i)}(\theta) = -\sum_c \mathbf{1}(y^{(i)} = c) \log p_\theta(y = c \mid x^{(i)}) = -\log p_\theta(y^{(i)} \mid x^{(i)})$; binary: $-y \log r - (1-y)\log(1-r)$ with $r = \sigma(s)$. True-class probabilities $0.9, 0.5, 0.1$ cost $0.105, 0.693, 2.303$: confident mistakes are punished hardest. **MLE.** If the labels are conditionally independent given the inputs, the probability the model gives to all observed labels is the *conditional likelihood* $L_{\text{cond}}(\theta) = \prod_i p_\theta(y^{(i)} \mid x^{(i)})$, and

$$-\log L_{\text{cond}}(\theta) = -\sum_{i=1}^{N} \log p_\theta(y^{(i)} \mid x^{(i)}) = N\, \widehat{R}_{\text{CE}}(\theta).$$

The log is increasing and $-1/N$ flips max into min without moving the optimiser: minimising empirical cross-entropy *is* maximising the conditional likelihood of the labels given the inputs, and nothing about $p(x)$ is ever modelled. **Gradient.** With $\frac{d}{ds}\log\sigma(s) = 1 - \sigma(s)$ and $\frac{d}{ds}\log(1 - \sigma(s)) = -\sigma(s)$,

$$\frac{\partial \ell}{\partial s} = -y(1-\sigma) + (1-y)\sigma = \sigma(s) - y, \qquad \nabla_w \ell = (\sigma(s) - y)\, x, \qquad \frac{\partial \ell}{\partial b} = \sigma(s) - y.$$

Prediction minus target, scaled by the input; the Hessian $\sigma(1-\sigma)\, xx^\top$ is positive semidefinite, so the loss is convex in (w, b). For softmax, $\partial \ell / \partial s_c = p_c - \mathbf{1}(y = c)$, the same form.

Key insight. The discriminative objective asks one question only: given x, which label? A large $\max_c p_\theta(y = c \mid x)$ means "this class beats the others on this input", never "this input looks like the training data" (§6).

Tutorial drill

Tutorial 1, Q6(a): *For $s = 1.2$ compute the posterior and, if $y = 1$, the cross-entropy.* $r_t = \sigma(1.2) = 1/(1 + e^{-1.2}) = 1/1.3012 = 0.7685$; $\ell_{\text{CE}} = -\ln 0.7685 = 0.263$ (equivalently $\ln(1 + e^{-s})$ when $y = 1$); had the label been 0 the cost would be $-\ln 0.2315 = 1.46$.
Tutorial 1, Q6(b): *Why is minimising empirical cross-entropy equivalent to maximising the conditional likelihood under an i.i.d. assumption?* The model's probability of the observed label is $r^y(1-r)^{1-y}$; independence of the labels given the inputs makes the probability of all N labels the product $L_{\text{cond}} = \prod_i r_i^{y_i}(1-r_i)^{1-y_i}$, whose log is $\sum_i [y_i \log r_i + (1-y_i)\log(1-r_i)] = -N\widehat{R}_{\text{CE}}$. Since log is increasing and $-1/N < 0$, $\arg\max L_{\text{cond}} = \arg\min \widehat{R}_{\text{CE}}$: logistic regression is maximum likelihood of $p(y \mid x)$ only; the inputs are conditioned on, never explained.
Tutorial 1, Q6(c)–(d): *$C_{\text{FP}} = 4$, $C_{\text{FN}} = 16$: threshold, decision for $s = 1.2$, and how the boundary of $s(x) = -3 + 2x_1 + x_2$ moves.* Threshold $t = 4/20 = 0.2$, in score space $s > \ln(0.2/0.8) = -1.386$; 1.2 qualifies, attack. At threshold 0.5 the boundary is $2x_1 + x_2 = 3$; at 0.2 it is $2x_1 + x_2 = 3 - 1.386 = 1.614$. Same normal vector $(2, 1)$, so the lines are parallel; the second is $1.386/\sqrt{5} = 0.62$ units closer to the origin and the attack half-plane has grown. Lowering the probability threshold means alerting on smaller scores, and "score above a smaller constant" is a half-plane bounded by a parallel line on the benign side: the direction is fixed by w, the cost only sets the constant.
New: *Two training examples both have label 1 and receive $\hat{p} = 0.818$ and $\hat{p} = 0.182$. Compute each cross-entropy and each gradient with respect to the score; which example drives the update?* Losses $-\ln 0.818 = 0.201$ and $-\ln 0.182 = 1.704$; gradients $\sigma - y = -0.182$ and -0.818: the confident mistake pulls four and a half times harder. Because $-\log p \to \infty$ as $p \to 0$, the fit can never "give up" on an example with a confident wrong probability, which is what calibration requires.

3 The Generative Route: Gaussian Class Models, LDA and QDA

The generative story and why its likelihood splits

A generative classifier imagines each example produced in two steps: draw a class $Y \sim \text{Categorical}(\pi)$, then draw features $X \mid (Y = c) \sim p_\theta(\cdot \mid y = c)$; parameters $\phi = (\pi, \theta)$. For independent examples the joint likelihood is $L(\phi) = \prod_i \pi_{y^{(i)}}\, p_\theta(x^{(i)} \mid y^{(i)})$, and grouping the terms by class,

$$-\log L(\phi) = \underbrace{-\sum_{c=1}^{C} N_c \log \pi_c}_{\text{prior part}} \underbrace{-\sum_{c=1}^{C} \sum_{i:\, y^{(i)}=c} \log p_\theta(x^{(i)} \mid y = c)}_{\text{one term per class}},$$

with N_c the number of examples of class c. The prior part depends on π only and each class term on that class's parameters only, so everything is fitted *separately*: $\hat{\pi}_c = N_c/N$ (Gibbs' inequality: the cross-entropy between the empirical class frequencies and π is minimal at equality), and the model of class c from the examples of class c alone. Each example is explained twice (class and features): more structure than the discriminative route, hence less data needed, hence more that can be wrong: a wrong family is a bias that persists with more data, finite-sample error shrinks.

Gaussian class-conditionals and their closed-form fit

$p_\theta(x \mid y = c) = \mathcal{N}(x; \mu_c, \Sigma_c) = (2\pi)^{-d/2}|\Sigma_c|^{-1/2} \exp[-\frac{1}{2}(x - \mu_c)^\top \Sigma_c^{-1}(x - \mu_c)]$. The mean μ_c is the centre of class c's cloud, the covariance Σ_c its shape (multiple of I: circles; unequal diagonal: axis-aligned ellipses; off-diagonal: rotated, correlated features); the quadratic form is the squared **Mahalanobis distance**, standard deviations from the centre in the direction of x. Maximum

likelihood is closed form, no iterations:

$$\hat{\pi}_c = \frac{N_c}{N}, \qquad \hat{\mu}_c = \frac{1}{N_c} \sum_{i:\, y^{(i)}=c} x^{(i)}, \qquad \widehat{\Sigma}_c = \frac{1}{N_c} \sum_{i:\, y^{(i)}=c} (x^{(i)} - \hat{\mu}_c)(x^{(i)} - \hat{\mu}_c)^\top.$$

The denominator is N_c (MLE), not the unbiased $N_c - 1$; say which one you use.

Generative scores; shared covariance gives LDA's linear log-odds; separate covariances give QDA

Define the **generative score** $s_c(x) = \log \hat{\pi}_c + \log p_{\hat{\theta}}(x \mid y = c)$. Then $\hat{\pi}_c\, p(x \mid c) = e^{s_c(x)}$ and Bayes' rule reads $p(y = c \mid x) = \mathrm{softmax}(s(x))_c$; for two classes $p(y = 1 \mid x) = \sigma(s_1(x) - s_0(x))$, so $s_1 - s_0$ is the log-odds.

LDA (shared Σ). Write the log density and expand the quadratic form, using the symmetry of Σ^{-1} to merge the two cross terms:

$$\log \mathcal{N}(x; \mu_c, \Sigma) = -\tfrac{d}{2} \log 2\pi - \tfrac{1}{2} \log |\Sigma| - \tfrac{1}{2} (x - \mu_c)^\top \Sigma^{-1} (x - \mu_c),$$
$$(x - \mu_c)^\top \Sigma^{-1} (x - \mu_c) = x^\top \Sigma^{-1} x - 2 x^\top \Sigma^{-1} \mu_c + \mu_c^\top \Sigma^{-1} \mu_c.$$

Because Σ is shared, $-\frac{d}{2} \log 2\pi$, $-\frac{1}{2} \log |\Sigma|$ and $-\frac{1}{2} x^\top \Sigma^{-1} x$ are identical for every class and drop out of any comparison. What remains is *affine in* x, and so are the log-odds:

$$s_c(x) = x^\top \Sigma^{-1} \mu_c - \tfrac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c,$$
$$\log \frac{p(1 \mid x)}{p(0 \mid x)} = s_1 - s_0 = \underbrace{(\mu_1 - \mu_0)^\top \Sigma^{-1}}_{w^\top} x \; \underbrace{- \tfrac{1}{2} \mu_1^\top \Sigma^{-1} \mu_1 + \tfrac{1}{2} \mu_0^\top \Sigma^{-1} \mu_0 + \log \tfrac{\pi_1}{\pi_0}}_{b}.$$

The boundary $s_1 = s_0$ is a hyperplane with normal $\Sigma^{-1}(\mu_1 - \mu_0)$; the prior enters only in b, so it *translates* the hyperplane and never rotates it. The posterior is exactly $\sigma(w^\top x + b)$: the logistic form, with w, b derived from $\hat{\mu}, \widehat{\Sigma}, \hat{\pi}$ instead of fitted directly. "Linear" describes the boundary, not the Gaussian.

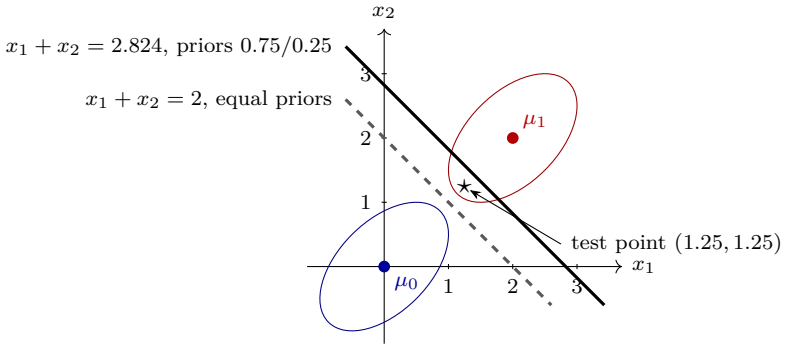
QDA (class-specific Σ_c). Now $\log |\Sigma_c|$ and $x^\top \Sigma_c^{-1} x$ depend on c and only $-\frac{d}{2} \log 2\pi$ cancels:

$$s_c(x) = -\tfrac{1}{2} (x - \mu_c)^\top \Sigma_c^{-1} (x - \mu_c) - \tfrac{1}{2} \log |\Sigma_c| + \log \pi_c.$$

Mahalanobis distance to the centre, a penalty for a diffuse class, the prior. $s_1 = s_0$ is quadratic in x: ellipses, parabolas, hyperbolas. In one dimension the x^2 coefficient is $\frac{1}{2\sigma_0^2} - \frac{1}{2\sigma_1^2}$, zero iff the variances are equal; otherwise two boundary points, and the wider class wins on both tails.

Tutorial 1, Q7 Part I: LDA with a correlated covariance

$\mu_0 = (0, 0)^\top$, $\mu_1 = (2, 2)^\top$, $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $\Sigma^{-1} = \frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$ (multiply out to check: I), priors 0.75/0.25. With $\rho = 0.5$: $\Sigma^{-1} = \begin{pmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{pmatrix}$. Class 0 ($\mu_0 = 0$): $s_0(x) = \ln 0.75 = -0.288$. Class 1: $\Sigma^{-1}\mu_1 = (4/3, 4/3)^\top$, so $x^\top \Sigma^{-1} \mu_1 = \frac{4}{3}(x_1 + x_2)$ and $\mu_1^\top \Sigma^{-1} \mu_1 = \frac{16}{3}$: $s_1(x) = \frac{4}{3}(x_1 + x_2) - \frac{8}{3} + \ln 0.25$. Boundary $s_1 = s_0$: $\frac{4}{3}(x_1 + x_2) = \frac{8}{3} + \ln 3$, i.e. $x_1 + x_2 = 2 + \frac{3}{4} \ln 3 = 2.824$. Test point $(1.25, 1.25)$: $x_1 + x_2 = 2.5 < 2.824$, class 0; $s_1 = 3.333 - 2.667 - 1.386 = -0.720$, $p(y = 1 \mid x) = e^{-0.720}/(e^{-0.288} + e^{-0.720}) = 0.487/(0.750 + 0.487) = 0.394$. Equal priors: the $\log \pi_c$ cancel, boundary $x_1 + x_2 = 2$ through the midpoint $(1, 1)$, test point now class 1 with $s_1 - s_0 = \frac{2}{3}$, $p = \sigma(0.667) = 0.661$.



One-standard-deviation contours ($\rho = 0.5$ stretches them along $(1, 1)$); the rarer class 1 must be "clearly closer" to win, so the line moves $0.824/\sqrt{2} = 0.58$ toward μ_1, past the test point.

Parameter counts: when LDA and QDA overfit

A symmetric $d \times d$ covariance has d diagonal plus $\binom{d}{2}$ off-diagonal entries, $d(d+1)/2$ numbers ($d = 100$: 5050). Priors cost $C - 1$, means Cd. LDA estimates one covariance, QDA one per class:

$$\text{LDA: } (C-1) + Cd + \tfrac{d(d+1)}{2}, \qquad \text{QDA: } (C-1) + Cd + C\,\tfrac{d(d+1)}{2}.$$

$d = 10$, $C = 2$: LDA $1 + 20 + 55 = 76$, QDA $1 + 20 + 110 = 131$. $d = 50$, $C = 3$: LDA $2 + 150 + 1275 = 1{,}427$, QDA $2 + 150 + 3825 = 3{,}977$. A class with $N_c \le d$ examples gives a covariance of rank at most $N_c - 1 < d$: singular, score undefined. QDA overfits first ($d(d+1)/2$ numbers *per class*); LDA pools all classes into one covariance. Remedy: **shrinkage** $\widehat{\Sigma}_\lambda = (1-\lambda)\widehat{\Sigma} + \lambda\alpha I$, positive definite for any $\lambda > 0$, λ chosen on validation data; at $\lambda \to 1$ all correlations are gone, which is the next section.

Tutorial drill

Tutorial 1, Q7 I(a): *Starting from the Gaussian log posterior, derive the LDA score and explain why comparing scores gives a linear boundary.* $\log p(c \mid x) = \log \pi_c + \log \mathcal{N}(x; \mu_c, \Sigma) - \log p(x)$, and $\log p(x)$ is class-free. Expand the quadratic form; $-\frac{d}{2}\log 2\pi$, $-\frac{1}{2}\log|\Sigma|$ and $-\frac{1}{2}x^\top\Sigma^{-1}x$ are identical for both classes *because* Σ is shared, so drop them: $s_c(x) = x^\top\Sigma^{-1}\mu_c - \frac{1}{2}\mu_c^\top\Sigma^{-1}\mu_c + \log\pi_c$, and $s_1 - s_0 = x^\top\Sigma^{-1}(\mu_1 - \mu_0) + \text{const}$ is linear, so its zero set is a line. The only term that could curve it cancelled.

Tutorial 1, Q7 I(d): *Explain how the prior moves the LDA boundary while the class models stay unchanged.* The prior appears only in the constant $\log(\pi_1/\pi_0)$ of the log-odds; the normal $\Sigma^{-1}(\mu_1 - \mu_0)$ comes from the class models alone. The line keeps its direction and translates from $x_1 + x_2 = 2$ to 2.824 when $\pi_1 = 0.25$: 0.58 toward the rare class, which must now be clearly closer to win. The point $(1.25, 1.25)$ lies between the lines and flips from class 1 (0.661) to class 0 (0.394).

New: $d = 10$ *features,* $C = 2$, $N = 60$ *examples. Count the parameters of LDA and QDA; which would you fit, and why?* LDA 76, QDA 131 (above). QDA estimates each 10×10 covariance from about 30 points: 55 numbers from 30 vectors is noisy at best and singular if a class has ≤ 10 points. Fit LDA (one covariance from all 60) or naive Bayes (10 variances per class). Flexibility the data cannot pay for becomes variance, not accuracy.

4 Naive Bayes

Fitting a full d-dimensional class-conditional is hopeless when d is large or the features are discrete with many levels: a covariance alone costs $d(d+1)/2$ numbers per class, and a joint table over d categorical features has exponentially many cells. Naive Bayes keeps the generative route and pays for it with one strong assumption.

Conditional independence and the additive score

Naive Bayes assumes the features are independent *given the class*:

$$p_\theta(x \mid y = c) = \prod_{j=1}^{d} p_\theta(x_j \mid y = c), \qquad s_c(x) = \log \hat{\pi}_c + \sum_{j=1}^{d} \log p_{\hat{\theta}}(x_j \mid y = c).$$

Read the score as a ledger: prior evidence, then one line of evidence per feature, each from a *one-dimensional* model of that feature within class c (Gaussian for measurements, Bernoulli for indicators, multinomial for counts, a hybrid for mixed tables; the prior appears once, never once per feature). A d-dimensional density problem becomes d one-dimensional ones, each stable on little data.

Why a wrong posterior can still classify well

The decision needs only the *largest* score; getting the argmax right is weaker than getting every density right. The characteristic failure is **double counting**: if x_2 is a copy of x_1 with log-likelihood ratio Λ, the true log-odds are $\log \frac{\pi_1}{\pi_0} + \Lambda$ (the copy carries nothing new), while naive Bayes multiplies the two likelihoods and computes $\log \frac{\pi_1}{\pi_0} + 2\Lambda$ (m copies: $m\Lambda$). Both have the same sign whenever Λ dominates, so the *decision* survives; the *probability* is pushed toward the favoured side and is too extreme. Flow example (Tutorial 1, Q7 II): prior odds 0.1/0.9, port ratio $0.60/0.05 = 12$, duration ratio $0.70/0.20 = 3.5$: odds $\frac{1}{9} \times 12 \times 3.5 = 4.67$, $p = 0.824$. If duration were a copy of port, naive Bayes would give $\frac{1}{9} \times 144 = 16$, $p = 0.94$, where the truth is $\frac{1}{9} \times 12 = 1.33$, $p = 0.57$. Removing redundant features repairs calibration; it rarely changes the ranking.

Zero counts, Laplace smoothing, and working in log space

Multinomial model. Vocabulary of V words, document $x = (x_1, \ldots, x_V)$ counts, $\theta_{cj} = p(\text{token} = j \mid y = c)$: $p_\theta(x \mid y = c) \propto \prod_j \theta_{cj}^{x_j}$ (the multinomial coefficient depends on x only and cancels in Bayes' rule; $0^0 = 1$ for absent words). A repeated word is repeated evidence; order is ignored. **The zero problem.** The MLE $\hat{\theta}_{cj} = n_{cj} / \sum_\ell n_{c\ell}$ is 0 for any word unseen in class c, and one zero factor kills the product: a single word vetoes the class with certainty, or, if unseen in every class, leaves the posterior 0/0. **Additive (Laplace) smoothing** pretends every word was seen α extra times in every class:

$$\hat{\theta}_{cj} = \frac{n_{cj} + \alpha}{\sum_{\ell=1}^{V} n_{c\ell} + \alpha V}, \qquad p(x_j = v \mid y = c) = \frac{n_{cjv} + \alpha}{n_{cj} + \alpha K_j} \text{ (categorical feature, } K_j \text{ levels).}$$

Every estimate is strictly positive and they still sum to one; $\alpha \to 0$ recovers the counts, $\alpha \to \infty$ gives uniform $1/V$. A categorical value never seen in training (a new port) needs an *other* level created at preprocessing, counted among the K_j levels and receiving every unseen value; smoothing then gives it a small positive probability instead of a crash. **Log space.** Products of dozens of probabilities underflow: compute $s_c = \log \hat{\pi}_c + \sum_j x_j \log \hat{\theta}_{cj}$, compare scores, and only if a posterior is asked normalise with $\sigma(s_1 - s_0)$ or softmax (subtract the largest score first).

The spam table, worked

Seven training tokens per class over the vocabulary (free, win, prize, meeting, project, report):

	free	win	prize	meeting	project	report
spam	3	2	2	0	0	0
ham	0	0	0	3	2	2

Unsmoothed, "free meeting": $p(x \mid \text{spam}) \propto \frac{3}{7} \cdot \frac{0}{7} = 0 = p(x \mid \text{ham})$, posterior 0/0. **Laplace,**

$\alpha = 1$, $V = 6$: denominators $7+6 = 13$, $\hat{\theta}_{\text{spam}} = \frac{1}{13}(4,3,3,1,1,1)$, $\hat{\theta}_{\text{ham}} = \frac{1}{13}(1,1,1,4,3,3)$. Equal priors. "free meeting": $\frac{1}{2} \cdot \frac{4}{13} \cdot \frac{1}{13}$ against $\frac{1}{2} \cdot \frac{1}{13} \cdot \frac{4}{13}$, equal, posterior $\frac{1}{2}$. "free prize" in log space: $s_{\text{spam}} = \ln\frac{1}{2} + \ln\frac{4}{13} + \ln\frac{3}{13} = -0.693 - 1.179 - 1.466 = -3.338$; $s_{\text{ham}} = -0.693 - 2.565 - 2.565 = -5.823$; $p(\text{spam} \mid x) = \sigma(2.485) = 0.923 = 12/13$. "free free prize": the repeated word widens the gap by $\ln 4$, posterior $0.980 = 48/49$; twice the word, twice the evidence.

Tutorial drill

Tutorial 1, Q7 II(a): $p(x_1 = 1 \mid \textit{attack}) = 0.60$, $p(x_2 = 1 \mid \textit{attack}) = 0.70$, $\pi = 0.10$; *benign* 0.05, 0.20, 0.90. *Scores and attack posterior for* $x_1 = x_2 = 1$. Attack $0.10 \times 0.60 \times 0.70 = 0.042$; benign $0.90 \times 0.05 \times 0.20 = 0.009$; $p(\text{attack} \mid x) = 0.042/0.051 = 0.824$: likelihood ratios 12 and 3.5 multiply to 42 against prior odds of 9.

Tutorial 1, Q7 II(b)–(c): *State the independence assumption, why it may fail for flows, and how correlated features produce overconfidence.* $p(x_1, x_2 \mid c) = p(x_1 \mid c)\, p(x_2 \mid c)$: once the class is known, port 22 says nothing more about duration. An SSH brute-force attack produces many *short* connections *to port 22*, one behaviour causing both, so within the attack class they co-occur; benign port-22 sessions are long logins, a dependence the other way. Each factor is then counted as independent evidence; a copied feature doubles Λ (odds 16, $p = 0.94$, against the true 1.33, 0.57). Ranking may survive, calibration does not, and a cost threshold on 0.94 thresholds a fiction.

Tutorial 1, Q7 II(d): *A port value appears in validation but never in training. How do an "other" category and additive smoothing prevent a zero likelihood?* Unseen value: $\hat{p}(v \mid c) = 0$ in every class, every score has a factor 0, posterior 0/0 (or, if seen in one class only, that class wins with certainty on one value). Fix one: at preprocessing map rare ports to a level *other* that has counts and is one of the K_j levels; at test time map every unknown port to it. Fix two: $(n_{cjv} + \alpha)/(n_{cj} + \alpha K_j)$; with 50 attack flows, $K_j = 6$, $\alpha = 1$, a level never seen among attacks gets $1/56$ instead of 0: positive, small, no longer a veto.

New: *Binary features, independent given the class. Show that Bernoulli naive Bayes is a linear classifier and say what it does and does not share with logistic regression.* $\log p(x_j \mid c) = x_j \log\theta_{cj} + (1 - x_j)\log(1 - \theta_{cj})$; subtract class 0 from class 1, sum over j, add the prior ratio: log-odds $= \sum_j w_j x_j + b$ with $w_j = \log\frac{\theta_{1j}(1-\theta_{0j})}{\theta_{0j}(1-\theta_{1j})}$, so $p(1 \mid x) = \sigma(w^\top x + b)$. Same hyperplane family as logistic regression, weights counted per class under independence rather than fitted by conditional likelihood: stable with few examples, blind to the interactions logistic regression learns with many.

5 Comparing the Two Routes

Question	Discriminative	Generative
Modelled quantity	posterior $p_\theta(y \mid x)$, or a boundary	prior π_c and class model $p_\theta(x \mid y = c)$
Input assumptions	few: nothing about $p(x)$	strong: a family for each class
Data efficiency	needs more labels; wins at large N if flexible	few parameters per class; wins at small N if the family is roughly right
Misspecification	no family to be wrong about x	a wrong family is a bias that *does not vanish* with more data
Missing features	impute or retrain	marginalise $p(x_{\text{obs}} \mid c) = \int p(x \mid c)\, dx_{\text{miss}}$; naive Bayes drops the term
Prior change	correct the odds afterwards (needs calibration)	swap π; class models untouched
Plausibility of x, sampling	none: softmax is relative	$p(x \mid c)$ can flag unfamiliar inputs and generate

Generative structure reduces estimation variance: a restricted description of each class has few parameters and is fitted reliably from little data, *provided the assumptions are roughly right*; a wrong family is a bias that stays. A flexible discriminative model spends no capacity on aspects of X irrelevant to Y and eventually reaches lower error. Error against N therefore often shows the generative curve lower at small N and the discriminative one lower at large N, but the crossing point, and whether it exists, is not a law. Same data, two verdicts: with 200 labelled emails and a 10,000-word vocabulary, multinomial naive Bayes estimates $2 \times 10{,}000$ smoothed word probabilities from counts and works from day one, while logistic regression must place a 10,000-dimensional hyperplane from 200 points and overfits; with 2 million emails logistic regression learns "*free* is spammy unless *invoice* is nearby", an interaction independence forbids naive Bayes from ever representing.

Tutorial drill

Tutorial 1, Q7 II(e): *Give one reason to prefer the discriminative route and one to prefer the generative route for flow classification.* Discriminative: with many labelled flows and strongly correlated features (bytes, packets, duration), it models $p(y \mid x)$ directly, does not double-count correlated evidence, and wastes nothing on how flows are distributed. Generative: with few labelled attacks its structured class models are stable; the prior is an explicit parameter, so a change in attack prevalence is fixed by swapping π; and $p(x \mid y)$ can flag a flow that fits no known class (§6).

Quiz refresher, C5(e): *LDA and logistic regression both give linear boundaries. In what do they differ, and when would you expect each to win?* Same form $\sigma(w^\top x + b)$, different estimation: LDA derives $w = \Sigma^{-1}(\mu_1 - \mu_0)$ from $\hat{\mu}_c, \widehat{\Sigma}, \hat{\pi}$ fitted by joint likelihood; logistic regression fits w, b by conditional likelihood, assuming nothing about $p(x \mid y)$. LDA wins with little data when the classes are roughly Gaussian with equal covariance; logistic regression wins with more data or when that assumption fails (outliers, skew), since a misspecified LDA converges to the wrong hyperplane and never recovers.

New: *At test time one of the d features is missing. What can each route do?* Generative: use the marginal $p(x_{\text{obs}} \mid c) = \int p(x \mid c)\, dx_{\text{miss}}$; for a Gaussian, drop that row and column of μ_c, Σ_c; for naive Bayes, omit the factor; Bayes' rule then proceeds unchanged. Discriminative: $p_\theta(y \mid x)$ is defined only for complete x, so impute (and inherit the imputation's error) or train one model per missingness pattern. Missing data at deployment is one of the few arguments for the generative route that does not depend on sample size.

6 Class Imbalance, Prior Shift and Unfamiliar Inputs

Two situations are constantly confused: a class that is *rare*, and a class whose frequency *changes* between training and deployment. A third is lurking behind both: an input that belongs to no class the model has seen. One definition first. A classifier is **calibrated** if among cases given probability u about a fraction u are positive; calibration is about the meaning of the numbers, the cost threshold about the action taken from them.

Imbalance is not shift

Class imbalance with the same distribution at training and deployment is *not* an error: the fitted posterior *should* reflect the prevalence. The problem is estimation: with few minority examples the training objective is dominated by the majority and the boundary around the minority is poorly placed; **weighting or resampling** makes the optimiser take the minority seriously, at the price that the fitted posterior now belongs to the *reweighted* prior and is no longer calibrated for the real one. **Class-prior shift** is a change of the world between training (t) and deployment (d): $p_d(x \mid y = c) = p_t(x \mid y = c)$ but $\pi_{d,c} \neq \pi_{t,c}$ (attacks 20% of training data, 5% of traffic). Imbalance with enough data needs no correction; shift needs one however much data there is.

Correcting for prior shift: swap the prior, or reweight the odds

Generative route. Bayes' rule at deployment uses π_d and $p_d(x \mid y) = p_t(x \mid y)$, which is what the fitted class models estimate: keep $p_{\hat\theta}(x \mid y = c)$, replace the prior, $p_d(y = c \mid x) = \pi_{d,c}\, p_{\hat\theta}(x \mid c) / \sum_k \pi_{d,k}\, p_{\hat\theta}(x \mid k)$. Nothing is refitted. **Discriminative route** (binary, $r_t(x) = p_t(y = 1 \mid x)$ calibrated). Step 1, odds form of Bayes' rule at each time: $\frac{r_t}{1-r_t} = \frac{\pi_t}{1-\pi_t}\Lambda(x)$ and $\frac{r_d}{1-r_d} = \frac{\pi_d}{1-\pi_d}\Lambda(x)$. Step 2, Λ is the same at both times (the definition of prior shift); divide:

$$\frac{r_d}{1-r_d} = \frac{\pi_d/(1-\pi_d)}{\pi_t/(1-\pi_t)} \cdot \frac{r_t}{1-r_t}.$$

Step 3, odds A/B means probability $A/(A+B)$ with $A = \frac{\pi_d}{\pi_t} r_t$ and $B = \frac{1-\pi_d}{1-\pi_t}(1-r_t)$:

$$r_d(x) = \frac{(\pi_d/\pi_t)\, r_t(x)}{(\pi_d/\pi_t)\, r_t(x) + \frac{1-\pi_d}{1-\pi_t}\,[1 - r_t(x)]}.$$

Reweight each class by how much commoner it became, renormalise. The formula silently needs a *calibrated* r_t; a model trained on balanced data has effective prior $\pi_t = 0.5$ and must be corrected with that value. **Order:** correct the probability for the deployment prior, then threshold it with the costs.

Key insight. Three knobs that look alike and are not: a *cost threshold* acts on the action (needs costs); a *prior-shift correction* acts on the probability (needs the deployment prevalence); *reweighting or resampling* acts on the estimate (needs only a tiny N_1) and breaks calibration. Algebraically all three end as a threshold on r_t; conceptually they answer three questions (Tutorial 1 asks this twice: Q2(d), Q6(f)).

Unfamiliar inputs: confidence is not familiarity

$\max_c p_{\hat\theta}(y = c \mid x)$ says that one modelled class outranks the others on x; it says nothing about whether x resembles the training data, because the discriminative objective never asked. A cat/dog/horse network shown a toaster outputs scores $(4.0, 0.5, -1.0)$, softmax $(0.96, 0.03, 0.01)$: "cat, 96%"; a new attack mechanism gets the same treatment. A generative model owns the missing question. **Reject option**: pick $c^* = \arg\max_c p(y = c \mid x)$, then accept only if $\log p_{\hat\theta}(x \mid y = c^*) \geq \tau_{c^*}$, otherwise reject or send to review. Two questions instead of one: which class fits best (posterior), and does even the best fit well enough (likelihood). The thresholds τ_c are validated on data that reproduce the intended novelty (attack mechanisms held out of training); a low likelihood is evidence of incompatibility, not a detection guarantee. Modern systems combine the routes: a discriminative ranker, a generative or anomaly model as its plausibility check.

Tutorial drill

Tutorial 1, Q6(e): $\pi_t = 0.20$, $\pi_d = 0.05$, *calibrated* $r_t = 0.60$*: compute* r_d. $\pi_d/\pi_t = 0.25$, $(1-\pi_d)/(1-\pi_t) = 0.95/0.80 = 1.1875$: $r_d = \frac{0.25 \times 0.60}{0.25 \times 0.60 + 1.1875 \times 0.40} = \frac{0.15}{0.625} = 0.24$. Odds check: training odds 1.5 times prior-odds ratio $\frac{0.05/0.95}{0.20/0.80} = 0.2105$ gives 0.316, $r_d = 0.316/1.316 = 0.24$. At the cost threshold 0.2 of Q6(c) the flow still alerts; at 0.5 it would flip from attack to benign.
Tutorial 1, Q2(d) and Q6(f): *Why is changing the threshold to reflect error costs different from correcting for class imbalance, and from the prior-shift correction?* The cost threshold acts at the decision stage: it takes the posterior as correct and asks what to do with it ($p > 2/10$ when a miss costs 8 and a false alarm 2); it needs the costs. The prior-shift correction acts on the probability because the world changed: same likelihood ratio, rarer attacks, weaker evidence; it needs the deployment prevalence. An imbalance correction acts at the estimation stage because the minority has too few examples. A threshold cannot repair a badly estimated or wrongly priored posterior, and reweighting encodes nothing about costs.

Tutorial 1, Q6(g): *When the minority sample is very small, why may weighting or resampling help even with no prior shift?* With a handful of attacks the empirical risk is dominated by benign flows; the fit can put the boundary where the benign data like it and treat the attacks as noise. Upweighting makes the optimiser pay for each missed attack and improves the boundary and the ranking. Price: the probabilities now belong to an artificial prior (0.5 after balancing); before thresholding, recalibrate or correct them with the odds formula using the *effective* training prior.
New: *Equal priors,* $X \mid 0 \sim \mathcal{N}(0,1)$, $X \mid 1 \sim \mathcal{N}(4,1)$. *Compute the posterior at* $x = 9$*; then apply a reject rule that accepts* 99% *of each class's own points. Why can the softmax of a discriminative model never do this?* Log-odds $-\frac{(9-4)^2}{2} + \frac{9^2}{2} = 28$, posterior $1 - 7 \times 10^{-13}$. Accepting $|x - \mu_{c^*}| \le 2.576$ means $\tau = -\frac{1}{2}\ln 2\pi - \frac{2.576^2}{2} = -4.24$; at $x = 9$, $\ln p(9 \mid 1) = -0.92 - 12.5 = -13.4 < -4.24$: **rejected**, five standard deviations from the class it is "certain" of. At $x = 2$ the posterior is 0.5 and $\ln p(2 \mid c) = -2.92 \ge -4.24$: accepted but ambiguous. Two kinds of doubt: overlap shows in the posterior, unfamiliarity only in the likelihood. A softmax is a ratio of exponentiated scores, invariant to a common shift: all classes may fit terribly and the ratio still be 0.96, and there is no $p(x \mid c)$ to compare with anything.

The mistakes that cost points

- Reading a probability threshold as a change of geometry. The score family fixes the shape; costs translate the boundary to $s = \log(C_{\text{FP}}/C_{\text{FN}})$, parallel to the old one.
- Deriving LDA without saying *why* the terms cancel: $x^\top \Sigma^{-1} x$ and $\log|\Sigma|$ drop only because Σ is shared; with class-specific Σ_c they stay and the boundary is quadratic (QDA).
- Confusing the three knobs: a cost threshold changes the action, a prior-shift correction the probability, reweighting the estimate (and breaks calibration). Correct first, then threshold.
- Reading $\max_c p(y = c \mid x)$ as familiarity (softmax is relative); counting the prior once per feature in naive Bayes; normalising before smoothing (add α to every count and αV to the total, then divide).

Chapter 4 in five lines

1. The decision rule is known (Bayes, costs); the posterior is not. Discriminative: fit $p_\theta(y \mid x)$. Generative: fit π_c and $p_\theta(x \mid y = c)$, apply Bayes' rule; posterior odds = prior odds × likelihood ratio. Both end in a softmax of scores.
2. Logistic regression: log-odds $= w^\top x + b$, boundary $s = 0$ a hyperplane, costs slide it to $s = \log(C_{\text{FP}}/C_{\text{FN}})$; cross-entropy $= -\frac{1}{N}\log L_{\text{cond}}$, gradient $(\sigma - y)x$, convex; softmax is shift-invariant, $\sigma(s_1 - s_2)$ for two classes.
3. Gaussian classes: MLE N_c/N, class means, covariances over N_c. Shared Σ: $x^\top \Sigma^{-1} x$ cancels, hyperplane with normal $\Sigma^{-1}(\mu_1 - \mu_0)$, prior only translates it (LDA). Separate Σ_c: quadratic boundaries (QDA), $d(d+1)/2$ parameters per class.
4. Naive Bayes: independence given the class, additive score, one 1-D model per feature; wrong calibration but often right ranking (a duplicated feature doubles Λ); Laplace smoothing $(n + \alpha)/(n_c + \alpha V)$ kills the zero-count veto; compute in logs.
5. Imbalance is an estimation issue (reweighting helps the fit, breaks calibration); prior shift is a change of the world (swap π, or reweight the odds of r_t, then threshold). Confidence is not familiarity: only a likelihood $p(x \mid c)$ can reject the unfamiliar.